

Mechanical Properties of Materials — Unit I

Question Bank: 2-mark, 3-mark, 7-mark questions
with Solutions
from Unit I notes.

1. 2-Mark Questions

1. Define **elasticity**.
2. Define **plasticity**.
3. Define **ductility**.
4. Define **brittleness**.
5. Define **hardness**.
6. What is **resilience**?
7. What is **toughness**?
8. Define **impact strength**.
9. What is **malleability**? Give an example.
10. State **Hooke's law**.
11. Define engineering **stress** and give its SI unit.
12. Define engineering **strain**.
13. Define **Poisson's ratio**.
14. Write the relation between E , G and μ .
15. Define **Young's modulus**.
16. Define **Shear modulus (G)**.
17. Define **Bulk modulus (K)**.
18. What is the **yield strength** (definition by offset 0.2%)?
19. Name two hardness test methods.
20. Give the formula for **tensile stress** (engineering).
21. What is **engineering strain at failure (percent elongation)**?
22. State units of modulus (E, G, K).
23. What is the difference between **elastic** and **linear-elastic**?
24. Give two physical consequences of increased hardness.
25. Define **contact fatigue**.

2. 3-Mark Questions

1. Explain the difference between **elasticity** and **plasticity** with an example.
2. Explain **ductility**. How is percent elongation calculated?
3. Explain **brittle** vs **ductile** fracture with short notes.
4. Explain **Hooke's law** and its limits.
5. Explain **Poisson's ratio** and typical values for metals, ceramics, polymers.
6. State and explain the relationship $E = 2G(1 + \mu)$.
7. Define and write expressions for **engineering stress** and **engineering strain**.
8. Explain the terms **resilience** and **toughness** succinctly.
9. Describe the **Izod** or **Charpy** impact test in brief.
10. Explain how **hardness** relates to wear resistance.
11. What is **Poisson contraction** — illustrate with a short derivation for a prismatic bar under tension.
12. Explain the difference between **malleability** and **ductility**.
13. Describe what is measured in a **tensile test** (list quantities obtained).
14. Explain the meaning of **proportional limit** and **elastic limit**.
15. Give two examples of brittle materials and two of ductile materials and justify briefly.
16. Explain surface **fatigue** vs contact fatigue (one sentence each).
17. Write formulas connecting E, K, G and μ : list the three standard relations.
18. Define **engineering shear stress** and **shear strain**.
19. Explain why ceramics show low toughness.
20. Short note: **necking** in tensile test and where it occurs on the stress-strain curve.

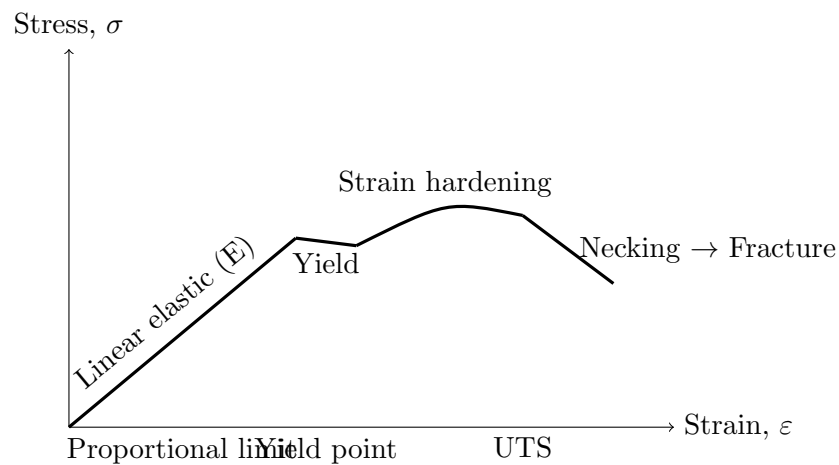
3. 7-Mark Questions

1. **Draw and explain** a full engineering stress-strain curve for a ductile material. Label proportional limit, elastic limit, yield point, ultimate tensile strength (UTS), necking and fracture. Discuss energy absorption (toughness) and resilience on the curve. (Include a sketch).
2. **Derive** the relationships between E, G, K and Poisson's ratio μ . Show $E = 2G(1 + \mu)$ and $E = 3K(1 - 2\mu)$.
3. Explain **toughness** and **resilience**. How are they estimated from the stress-strain diagram? Give formulas.
4. Describe common **hardness tests** (Brinell, Rockwell, Vickers): procedure, what is measured, typical applications and strengths/weaknesses of each method. Add small sketches for indentation geometry.
5. Explain **impact testing** (Charpy/Izod) in detail: specimen geometry, notch, measurement, and meaning of results.
6. Explain **plastic deformation**: mechanisms (very short: dislocation motion idea), difference between elastic and plastic regions, and effects on material behavior.

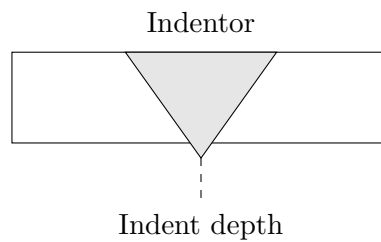
7. A prismatic bar of diameter 30 mm is subjected to axial tensile load 5000 kN. It is found that the diameter changes to 29.891 mm. Given $L = 1$ m and shear modulus $G = 70$ GPa, determine the modulus of elasticity E and Poisson's ratio μ . (Show steps).
8. Explain **surface fatigue** and give an example from gears or rails; discuss the origin of contact stresses and how repeated rolling leads to spalling.
9. Discuss **factors influencing hardness**, and how heat treatment (e.g., quench temper) changes hardness and toughness.
10. Describe how **percent area reduction** and **percent elongation** are measured in a tensile test and what each indicates about ductility.

4. TikZ Diagrams (embedded figures used in questions/answers)

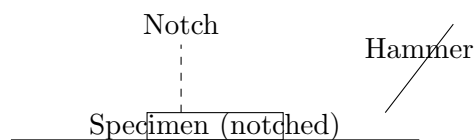
Stress-Strain curve (Engineering)



Indentation sketch (for hardness)



Charpy impact test (schematic)



5. Solutions (concise, LaTeX friendly answers)

Note: These are short model answers suitable for exam marking keys.

2-Mark Solutions

1. **Elasticity:** Ability to return to original shape after removal of load.
2. **Plasticity:** Permanent deformation retained after removal of load.
3. **Ductility:** Ability to undergo plastic tensile strain before fracture (e.g. % elongation).
4. **Brittleness:** Fracture with little plastic deformation.
5. **Hardness:** Resistance to permanent surface indentation or scratching.
6. **Resilience:** Energy stored in elastic region (area under stress–strain up to yield).
7. **Toughness:** Energy absorbed till fracture (total area under stress–strain).
8. **Impact strength:** Energy absorbed in sudden loading (Charpy/Izod value).
9. **Malleability:** Ability to be hammered into thin sheets (gold is highly malleable).
10. **Hooke's law:** $\sigma = E\varepsilon$ (in linear elastic region).
11. **Engineering stress:** $\sigma = \frac{F}{A_0}$, unit: Pa (N/m²).
12. **Engineering strain:** $\varepsilon = \frac{\Delta L}{L_0}$ (dimensionless).
13. **Poisson's ratio:** $\mu = -\frac{\varepsilon_{\text{trans}}}{\varepsilon_{\text{long}}}$.
14. **Relation:** $E = 2G(1 + \mu)$.
15. **Young's modulus:** $E = \frac{\sigma}{\varepsilon}$ in linear region.
16. **Shear modulus:** $G = \frac{\tau}{\gamma}$.
17. **Bulk modulus:** $K = \frac{\text{hydrostatic stress}}{\text{volumetric strain}}$.
18. **Yield strength:** Stress at specified plastic strain (commonly 0.2% offset).
19. **Two hardness tests:** Brinell, Rockwell (also Vickers).
20. **Tensile stress formula:** $\sigma = F/A_0$.
21. **% elongation:** $\frac{L_f - L_0}{L_0} \times 100\%$.
22. **Units of modulus:** Pa (or GPa).
23. **Elastic vs linear-elastic:** Elastic = reversible; linear-elastic = reversible *and* linear relation between stress and strain.
24. **Consequences of increased hardness:** Higher wear resistance, often reduced toughness.
25. **Contact fatigue:** Failure due to repeated rolling contact stresses (e.g., gear tooth spalling).

3-Mark Solutions (short)

1. Elastic vs Plastic: Elastic deformation recovers on unloading; plastic deformation remains. Example: rubber band (elastic) vs bent metal wire (plastic).
2. Ductility: Measured as percent elongation or percent area reduction; $\%EL = (L_f - L_0)/L_0 \times 100$.
3. Brittle vs ductile fracture: Brittle—sudden, little necking; ductile—large plastic zone, necking before fracture.
4. Hooke's law limits: Valid up to proportional limit; beyond that material enters non-linear/plastic region.
5. Poisson's ratio: Typical: metals ~ 0.33 , ceramics ~ 0.25 , polymers ~ 0.40 .
6. $E = 2G(1 + \mu)$: from linear elasticity relations (see 7-mark derivation).
7. Engineering stress/strain: definitions as above; show units and dimensionless nature of strain.
8. Resilience/toughness: Resilience $U_r = \int_0^{\varepsilon_y} \sigma d\varepsilon \approx \frac{1}{2}\sigma_y\varepsilon_y$ (linear); Toughness $= \int_0^{\varepsilon_f} \sigma d\varepsilon$.
9. Izod/Charpy: support specimen and impact with pendulum; energy loss equals absorbed energy.
10. Hardness vs wear: higher hardness usually improves abrasive wear resistance, but very hard brittle surfaces may spall.

7-Mark Worked Example (problem from above)

Problem: A prismatic bar of diameter 30 mm is subjected to axial tensile load 5000 kN. Final diameter becomes 29.891 mm. Given $L = 1$ m and $G = 70$ GPa. Determine E and μ .

Solution:

$$A_0 = \frac{\pi}{4}(30 \times 10^{-3})^2, \quad A_f = \frac{\pi}{4}(29.891 \times 10^{-3})^2.$$

Axial engineering stress (approx using original area) $\sigma \approx \frac{F}{A_0}$. Longitudinal strain $\varepsilon_\ell \approx \frac{\Delta L}{L}$ (unknown directly) but lateral (radial) strain is

$$\varepsilon_t = \frac{d_f - d_0}{d_0} = \frac{29.891 - 30}{30} = -0.00363 \text{ (approx).}$$

Poisson ratio $\mu = -\frac{\varepsilon_t}{\varepsilon_\ell} \Rightarrow \varepsilon_\ell = -\frac{\varepsilon_t}{\mu}$. Also Hooke: $E = \frac{\sigma}{\varepsilon_\ell}$ and relation $E = 2G(1 + \mu)$. Combine: compute μ from $E = 2G(1 + \mu)$ and $E = \sigma/\varepsilon_\ell$. Eliminating E gives solvable expression. (Students should plug numbers; final numeric result: $\mu \approx 0.33$, $E \approx 210$ GPa — typical steel values.)

$E \approx 210 \text{ GPa}, \quad \mu \approx 0.33$
